

Practice Quiz 01B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Truth tables and logical equivalence.

1. Use a complete truth table to determine whether

$$P \rightarrow Q \quad \text{and} \quad \neg P \vee Q$$

are logically equivalent. State your conclusion.

Solution

P	Q	$P \rightarrow Q$	$\neg P \vee Q$
F	F	T	T
F	T	T	T
T	F	F	F
T	T	T	T

The two columns agree in every row, so the formulas are logically equivalent. Thus a conditional can be rewritten as $P \rightarrow Q \equiv \neg P \vee Q$. ■